CS-011 Inventory, Storage & Transport

Doc ID	CS-011
Title	Inventory, Storage & Transport
Revision	D
Status	Draft, pending Lead signature (storage-map locations to field-verify)
Effective date	2026-08-28
Supersedes	The stale Master Tracker Inventory sheet as a live record

Approvals

Role	Name	Date
Author	Simon Starbuck (drafted with Claude, SN6 Resources)	2026-07-12
Reviewer	simon (reviewer agent)	
Approver (Composites Lead)		

Revision history

Rev	Date	Author	Description of change
A	2026-07-12	Claude	Initial outline
B	2026-07-28	Claude	Added a reference to the app's Stock tab and noted its location field is freeform today, not a controlled vocabulary, so it doesn't yet enforce the storage map. Language pass.
C	2026-08-03	Claude	§7.3 storage map: locations become records with their own labels, so the location field is a controlled value rather than freeform text (CS-001 §7.10).
D	2026-08-28	Claude	Engineering pass (CS-000 Rev C conventions). §7 dropped its "(outline)" label and its five run-on items became titled subsections; the storage-map paragraph was split so the rule, the mechanism and the payoff read separately. No rule changed.

1. Purpose

Our SN5 inventory sheet was written at season start and then never touched. MEKP sat flagged "RE-ORDER" for months with no owner and no trigger, and when the October Easy Composites order got stuck in customs we had no buffer and lost weeks of layups (PP-02). Storage was purely reactive: parts got evicted from the etch locker, molds blocked everyone's boxes at RFS (PP-10). This doc sets minimum stocks, gives them an owner, and maps where things live.

2. Scope

Consumables, resins/chemicals, cloth, tooling board, molds, and finished parts: where they live, when they get reordered, how they move.

3. References

Ref	Document	Where
R1	PP-02, PP-10 RCAs	FEB-COMP-PP-001
R2	CS-012 (ordering process + lead times)	CS-012
R3	CS-001 (labels carry home locations)	CS-001
R4	SDS chemical storage sections	`03 Datasheets/`
R5	The app's Stock tab (live tooling-board + offcut inventory)	`06 Composites App/app/README.md`

4. Definitions

Min-stock (the level that triggers a reorder, sized so the order lands before stock-out at the supplier's lead time), stock walk, home location.

5. Equipment & Materials: layup-blocker min-stock list (starting values; tune with usage data)

Sizing rule (§4): min stock = (burn rate × supplier lead time) + one job's worth. Worked examples for the three items that actually stop a layup; the rest of the table starts as one-job buffers, tuned with WO usage data:

- **IN2 + AT30:** SN5 logged ~1.95 kg mixed per large infusion; at the 2-infusions/week peak that is ~4 kg/week. Easy Composites lead 6 weeks (CS-012) → running dry mid-season costs ~6 weeks of infusions. One unopened kit (~4 kg) 1 peak week: it is the *alarm*, not the buffer: when the walk finds only the unopened kit, the order is already late, so trigger at **opened kit + 1 unopened**, and place the big seasonal orders per CS-012 §7.4 so the walk never trips this.
- **XCR:** ~1 kit seals 2–3 mold sections (tune from WO records); molds queue 2/week in peak → 1 kit 1 week. Same logic: **1 unopened kit minimum**, ordered with every EC basket.
- **Tacky tape:** every bagged job uses 1–2 rolls of perimeter; 2 jobs/week → ~3 rolls/week; domestic lead ~1 week → **6 rolls** 2 weeks of peak burn.

Item	Min stock	Why it blocks
IN2 resin + AT30	1 unopened kit (+ the opened one)	Every infusion; 6-week EC lead
West 105 + 206	½ tank + 1 can	Every wet layup
XCR	1 kit	Every mold
VB160 bag film	1 roll	Everything bagged

Item	Min stock	Why it blocks
Peel ply / flow mesh	1 roll each	Everything bagged
Tacky tape	6 rolls	Everything bagged; team burns it fast
Nitrile gloves	2 boxes/size	Everything
Mixing cups/sticks	50/50	Everything
Respirator cartridges (A + P100)	2 sets	Safety-blocking
195 twill cloth	(sponsor) 10 yd	Most stacks

6. Safety

Chemical storage per each SDS (R4): resins/hardeners in the chemical boxes/barrel, **never in food fridges** (2026-02-17 incident); flammables (Airtac, acetone, clear coat) in the flammables cabinet; segregate hardeners from resins on different shelves.

7. Procedure

7.1 Monthly stock walk

1. 15 minutes, named owner; an IA or first-year is the right owner, it is a good first responsibility.
2. Check the §5 list against reality. Anything at or below min goes to #purchasing the same day, per CS-012.

7.2 Order timing

1. Easy Composites orders land 6 weeks before need (UK shipping plus customs, PP-02).
2. One big fall order plus one spring top-up beats four rushed baskets; the calendar is CS-012 §7.4.

7.3 Storage map

1. Home locations (field-verify on first revision): molds → RFS storage container, never in front of the knaack or screw boxes; finished parts → Jacobs basement shelves, never the etch locker (2026-04-20 agreement); cloth → dry sealed bin; chemicals → per §6; consumables → General Box at RFS.
2. Board and offcuts are logged in the app's Stock tab (R5), which is where real inventory lives day to day.
3. Every shelf, bin and rack gets a storage record and a printed label on its front edge (CS-001 §7.10), and every mold, part and material lot points at one. Recording a move is: scan the object, tap Move, scan the shelf.
4. The payoff, enforceable since 2026-08-03: "where is the seat mold" is a lookup rather than a Slack ask, and the monthly walk checks reality against a controlled value instead of whatever string somebody typed.

7.4 Transport

1. Molds over ~1 m^2: two people, and a vehicle that actually fits them. The SN5 budget shows a U-Haul was needed; plan it, don't discover it.
2. Parts travel padded and labeled.

7.5 Season handoff

1. At season end, the stock-walk owner and the lead produce a one-page "what we actually have" list for the incoming lead (the 6/15 handoff gap).

8. Quality checks / acceptance criteria

Check	Target	Method	Record where
Stock walk done	Monthly during build season	Owner posts result in #composites	Slack + walk sheet
No layup blocked by stock-out	0 per season	WO delay notes	WO timeline
Molds in home locations	100% at walk	Walk	Walk sheet + WO mold.location

9. Records

Walk sheets (photo → Drive), WO mold.location fields, reorder requests in #purchasing.